

Lokesh Poluru Velayudham

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SUMMARY

Senior Software Engineer specializing in AI-driven medical imaging systems and large-scale inference pipelines. Experienced in building GPU-accelerated ML systems, full-stack cloud platforms, and distributed data pipelines for clinical applications, including systems submitted for FDA clearance. Strong background in Python, TypeScript, and cloud-native architectures with a focus on performance, scalability, and reliability.

SKILLS

- **Programming & Frameworks:** Python, TypeScript, JavaScript, C++, Node.js, NextJS, FastAPI, Java, Spring Boot, C#, C (STL, multithreading)
- **ML & AI Integration:** PyTorch, Transformers (ViT, BERT, CLIP), Generative AI, LLM APIs, Model Serving, Data Enrichment Pipelines
- **ML Systems & Inference:** Batch & real-time inference, Model optimization (FP16), Request batching / async, deployment & serving (REST/gRPC)
- **Architecture & Systems Design:** Microservices, Service-Oriented Architecture (SOA), Event-Driven Design, Distributed Systems, Scalability
- **Data & Infrastructure:** PostgreSQL, MySQL, MongoDB, Redis, Kafka, Spark, Flink, DynamoDB, Data Pipelines (Streaming & Batch), ETL Optimize
- **Cloud & DevOps:** AWS (EC2, S3, Lambda, CloudFormation, CloudWatch), Docker, Kubernetes, Terraform, CI/CD, Monitoring & Observability
- **API & Integrations:** RESTful Services, GraphQL, gRPC, WebSockets, OAuth2/JWT, API Gateway, Backend-for-Frontend (BFF)
- **Frontend Engineering:** React, Next.js, TypeScript, WebGL, vtk.js, Responsive Design, Qt/QML, Real-Time Data Visualization

EXPERIENCE

Software Engineer – Medical Imaging & AI Systems

BioInVision Inc

January 2025 – Present, Cleveland, OH

- **StainViz (Virtual Staining AI):** Built GPU-accelerated inference pipelines using GANs and ViT models in PyTorch, achieving **40% latency reduction** via FP16 optimization, batching, and memory tuning, deployed in production-grade clinical workflows.
- **IschemiaViz (CCTP – Cardiac CT Perfusion):** Engineered a clinical-grade medical imaging system submitted for **FDA clearance under QMS processes**, delivering **<2s inference on 10GB datasets** using multithreaded C++/Python pipelines with reproducible and improved diagnostic.
- **CryoViz Web (3D Microscopy Platform):** Developed a full-stack cloud platform (React, Next.js, Azure/AWS) for **microscopic-resolution, single-cell sensitivity imaging of mouse-sized specimens**, enabling sub-second slice retrieval for 800MB+ datasets via chunked streaming and vtk.js rendering.
- **Backend, Data & Distributed Systems:** Designed scalable backend services and **database schemas (PostgreSQL/MongoDB)** for large-scale imaging data, and annotation workflows; built distributed semantic search and vector-based retrieval systems for efficient navigation across 3D datasets.
- **MLOps, APIs & System Performance:** Implemented high-throughput ingestion pipelines (async Python, memory-mapped I/O), reducing load times to **<15s**; containerized AI inference services using Docker and deployed via REST/gRPC with observability (logging/monitoring), ensuring scalable, reproducible, and reliable production workflows.

Software Developer | RA | TA – AI & Full-Stack Systems

Case Western Reserve University – Undergraduate Research Office

January 2024 – January 2025, Cleveland, OH

- Containerized **GPU-accelerated Pix2Pix GAN** pipelines with **Docker**, standardizing mixed-precision (FP16/FP32) execution and enabling portable, reproducible deployment across environments while reducing setup time and ensuring consistent high-performance inference.
- Applied **AI3** principles to redesign model execution on GPU, tuning precision scaling, kernel utilization, and data movement to **minimize memory** bottlenecks and accelerate **end-to-end inference** performance for high-resolution image processing workloads.
- Owned design of an **AI assisted** judge and student matching platform on a MERN stack, pairing **1,500** students with **250** judges and cutting setup time from days to **30** minutes using **ML** based match scoring.

Backend Engineer – Mobility SaaS Platform

Alter Cabs

June 2023 – January 2024, Chennai, India

- Architected and delivered a ride hailing SaaS backend with a React frontend, scaling to **4,000** drivers with high availability and producing operational data suitable for **time series forecasting**.
- Developed and launched the Quick Ride dispatch feature cutting ride confirmation time to under 5 minutes through optimized matching algorithms, generating streams that enabled **SSE** and **WebSockets** for demand forecasting and driver allocation.
- Implemented Docker based **CI and CD** pipelines on AWS that improved deployment reliability and reduced rollback incidents 35%, laying the foundation for **CI and CD for AI systems** and robust **observability** across production services.

Software Engineer – B2B SaaS (HORECA Domain) – (Founding Member)

Foxx Technologies

June 2019 – May 2023, Chennai, India

- Designed backend services for a B2B SaaS ordering platform in **SpringBoot (Java)** and **React** serving **50** enterprise clients with high concurrency and fault tolerance, instrumented for integration with **Azure Databricks** and **vector database** powered recommendation.
- Engineered **Firebase** and **Golang** based microservices that boosted search performance 30% and enabled consistent sub second responses, while shaping schemas compatible with **embeddings** and **semantic search**.
- Directed MVP delivery ahead of schedule, helping secure **4.5M** rupees in funding and generate 0.75 crore revenue in 6 months, using data metrics that later supported **forecasting** of growth and retention.

Software Developer – EdTech Systems

Guvi Geek Technologies (IIT-Madras Incubated)

January 2018 – June 2019, Chennai, India

- Owned development of a **Java** based hiring platform module connecting **50** employers with **1,000** students in **4** months, streamlining workflows.
- Integrated a microservices architecture that improved system performance **25%** and enhanced scalability for growing employer and student activity.

EDUCATION

Master of Science in Computer and Data Science

Case Western Reserve University • Cleveland, Ohio • 2025 • 3.9